Cellular Automata for Two Lane Traffic Flow

Abubakar Othman (F221611)

Mathematical Modelling II (25MAP211)

Department of Mathematical Sciences

Abstract

Key Words:

Contents

1 Introduction

1.1 Real-world Context

Urban Traffic Congestion is a pervasive issues across a range of different demographics, whether it is in a developing or developed country. Traffic Congestion has always been a significant concern in traffic control mechanisms caused by an unbalanced demand-supply situation that directly affects the problem of large fuel consumption and greenhouse emission. Congestion leads can lead to a range of issues such as traffic jams, unhealthy air quality, vibrations, noises and accidents which are significant problems for the environment and human health [1].

In the UK, this issue is particularly prevalent. In the 2019, UK drivers lost an average of 115 hours each to road traffic congestion [2].

To reduce congestion, we should look at optimising traffic flow, this can be done through the use of traffic lights, by employing various strategies we can maximise the flow of traffic and reduce congestion.

1.2 Why discrete cellular automata

We can leverage the use of cellular automata to model traffic flow, this is because they are a powerful tool for simulating complex systems and can capture the dynamics of traffic flow in a simple and efficient way. Cellular Automata also allow us to assess the impact of different traffic light strategies on traffic flow, and can help us to identify the most effective strategies for reducing congestion.

1.3 Aims and Research Questions

1.4 Structure of The Report

2 Background

3 Model Description and Boundary Form

3.1 Model Assumptions

3.2 Spatial Domain and Policy Setup

4 Numerical Method and Validation

4.1 Numerical Implementation

4.2 Validation

5 Baseline Results

6 Sensitivity Analysis

7 Conclusion

References

[1] U. Jilani, M. Asif, M. Y. I. Zia, M. Rashid, S. Shams, and P. Otero, "A Systematic Review on Urban Road Traffic Congestion," *Wireless Personal Communications*, vol. 140, pp. 81–109, Jan. 2025.

[2] B. Pishue, "2020 INRIX Global Traffic Scorecard,"